

Deepak Mallampati

APPLIED MACHINE LEARNING ENGINEER

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Kent, OH - F-1 OPT (authorized now); open to Chicago relocation

PROFESSIONAL SUMMARY

Applied ML engineer with 4+ years translating research into production for regulated healthcare and financial systems. Built a privacy-preserving clinical ML pipeline (re-identification risk reduced 3.38% → 0.32% while retaining 99% of baseline predictive performance) and authored an IEEE-format paper on the methodology. Ship regression, embedding, and retrieval models end to end — production RAG at 92% precision with 40% hallucination reduction over millions of documents — with the MLOps discipline (CI/CD, monitoring, evaluation) to keep them reliable in production.

CORE COMPETENCIES

Applied Machine Learning • Regression & Classification • Embeddings & Retrieval • Research → Production • Probabilistic / Privacy Modeling • MLOps & Model Monitoring • Healthcare Analytics

ML & Modeling: Python, PyTorch, TensorFlow, scikit-learn | Regression, Classification, Embeddings, Clustering | NumPy, Pandas | Differential Privacy (Laplace), k-anonymity / l-diversity | RAGAS, BERTScore

Retrieval & LLMs: FAISS, Pinecone, Weaviate, pgvector | Dense retrieval + cross-encoder re-ranking | LangChain/LangGraph | Fine-tuning (LoRA, QLoRA, PEFT)

Infra & Cloud: AWS (SageMaker, Bedrock, S3, Lambda), Azure, GCP Vertex AI | Docker, Kubernetes, Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | PostgreSQL, Spark/Databricks

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI / Machine Learning Engineer

- Engineered high-performance **retrieval-augmented (embedding-based) pipelines** over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination by **40%** at **92% precision**.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (k-anonymity, l-diversity, Laplace differential privacy), enabling compliant analytics and model development with zero PII exposure.
- Architected **agentic LLM systems** on a conductor-subagent framework, cutting token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Established a structured **ML evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark iterations and catch regressions before release.
- Built fault-tolerant automation (scheduled orchestration, exponential-backoff retries), reducing failed production runs by **60%**.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset; reduced re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696).
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations and quantifying differential privacy's utility cost across four ϵ levels — a probabilistic trade-off analysis directly analogous to causal/Bayesian modeling.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100; authored a peer-review-ready **IEEE-format conference paper** documenting methodology and results.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering AI capabilities, reducing p95 latency 1.5s → **800ms** and supporting **100,000+ requests daily** across 20+ teams.
- Evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates by **27%** on internal evaluation datasets and informing model-selection decisions.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Developed and trained supervised **regression and classification models** on operational data for equipment-failure prediction, anomaly detection, and capacity forecasting, with end-to-end feature engineering and preprocessing.
- Fine-tuned BERT/RoBERTa for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

Framework for analyzing sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy — quantifying exactly how much privacy each technique buys for how little accuracy cost.

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

EDUCATION

M.S. in Computer Science, Kent State University, OH

2025 | GPA 3.70/4.0